Week 2 teaching note: The Finance View

MSc Finance: The Future of Capitalism

Prof.dr. Arjen Siegmann

8 Feb. 2026

Introduction

On a recent Wednesday I (Arjen) stepped into the elevator and met Jos, a colleague from spatial economics. He wondered why, in Finance, we never seem to study welfare. "In my field, this is all we do, but you Finance people never seem to care." In a nutshell, the story illustrates the need for this course. We should study how Finance can benefit society. How we can mitigate the downsides. And how the Finance-perspective can help us to better understand the world around us.

Last week, we dealt with Marx' view on society, historical materialism. But something seems to be missing: why is there so much growth? Why has poverty gone down tremendously in the past century? And, what does Finance contribute to this?

This week we take the Finance view and answer these questions. The answers are not perfect nor comprehensive, i.e., there is much to say about it. But, it does give a good solid core for thinking about Finance and society.

What explains our current standard of living and level of technology? Our workhorse model is that of a balance sheet of a corporation. In one sentence: Value is created by putting assets to work, in a corporate structure that enables risk, that can be diversified by investors and debt funded by banks that transform near risk-free short-term savings into long-term loans.

The note has four parts. Part 1 is about value creation through capital budgeting. Part 2 is about how corporations have become a vehicle for exploration and risk taking. Financial option theory is used to show how this works. Part 3 is about the power of diversification, as shown by the CAPM model. Part 4 is about the role of banks in society using the Diamond-Dybvig model. From that model, we can directly derive the need for financial regulation. This regulation is needed to have the benefits that banks bring, without suffering from too many crises. In a sense, deposit insurance is one answer to Marx.

1 Value creation through capital budgeting

We live in a world of extraordinary material abundance compared to the world Marx inhabited. Pick up almost any metric of human welfare—life expectancy, infant mortality, nutrition, access to education, housing standards—and the same pattern emerges: dramatic improvement, especially in the last two centuries. This improvement is not uniform across countries, nor is it steady within countries, but the long-run trajectory is unmistakable.

Consider child mortality, perhaps the most fundamental measure of a society's ability to sustain life. Around 1800, roughly 40-50% of all children born did not survive to age five. This was not exceptional; it was normal. By 1950, this had fallen to roughly 20%. By 2020, it stands below 5% globally, and in wealthy countries, below 1%. Over two centuries, child mortality has dropped by an order of magnitude.[1] Figure 1 shows this dramatic decline.

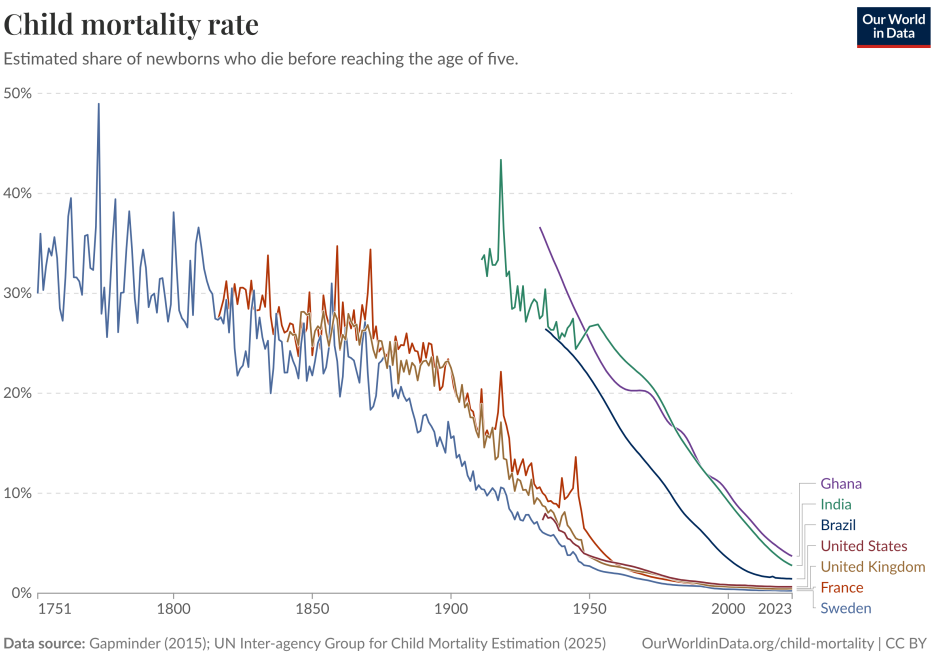


Figure 1: Global Child Mortality Rate (deaths per 100 live births), 1751–2023. Source: Gapminder and UN IGME.

This is progress on a scale that demands explanation. Marx was writing in the 1850s-1880s, when childhood death was still the norm for most of humanity. He could observe the destructive side of capitalism—the exploitation of workers, the precarious conditions in factories, the concentration of wealth—but he could not observe what we now know: that capitalist economies would eventually generate enough wealth and organizational capacity to save the lives of billions of children. The question is: where does this wealth come from, and how is it created?

The standard Marxist answer to growth is insufficient. Marx understood capitalism as a sys-

[1] Data from Gapminder and UN Inter-agency Group for Child Mortality Estimation (IGME), compiled by Johansson and Lindgren. See Gapminder (2015) and UNIGME (2024).

tem of extraction: the capitalist class appropriates the surplus value created by workers' labour. Competition drives accumulation, but the tendency is toward crisis, not sustainable expansion. Growth, in this framework, is derivative: it comes from expansion of the labour force, deeper exploitation, or conquest of new markets. But it does not come from systematic innovation, improvement of technique, or the creation of new products and services.

What Marx did not fully theorize—what would not become clear until much later—is that the "capitalist mode of production" (Dutch: "ondernemingsgewijze productie") has a mechanism for generating *innovation and exploration*.

First step is the balance sheet of a corporation.

Assets (A)	**Equity (E)**
	Debt (D)

How do the assets grow of a corporation? Because capital budgeting takes place. The manager allocates capital to projects based on their cash flow profile, as you learn in *Corporate Finance*. Projects are selected based on the highest *Net Present Value*.

Say we are given an investment opportunity of US$ 100 mln, with a cash flow of US$ 121 at the end of year 1. With a discount rate of r, we would value this project as follows:

$$NPV = -100 + 121/(1 + r) \tag{1}$$

When do we do this project? When the discount rate r is smaller than 21%. If that is the case, the NPV is positive and we are creating value. (Well, we do if there are no other projects with a better NPV.) Value is created *at the moment* we take this decision. Because the 121 payoff is uncertain and we never know what will happen. But by being disciplined and using the NPV-criterion, in expectation, we will be creating value for the firm.

This little example is enough to understand why economic growth keeps going on: as long as entrepreneurs do proper capital budgeting, they keep selecting projects that beat the hurdle rate of r, the cost of capital. This creates value and lets the economy grow, in the aggregate. Some projects fail, others succeed, but the average should be good.

2 The power of risk (Equity as an option)

But who can bear the risk of value creation? The mechanism critically critically on a legal and financial innovation that was still new in Marx's time: the joint-stock company with

limited liability. Understanding this innovation is the key to understanding where sustained growth comes from.

2.1 The Modern Corporation: A Legal Invention

The corporation is not a law of nature. It is a legal fiction, invented to solve a coordination problem. Before the joint-stock company, most enterprise was organized as a partnership or family firm. If you wanted to finance a large expedition or a risky venture, you pooled capital with partners, and each partner was liable—personally and fully—for all debts of the enterprise. If the venture failed spectacularly, you could lose not just your investment but your home, your land, everything.

This structure worked reasonably well for small-scale, low-risk ventures. It worked poorly for exploration, colonization, and long-distance trade, which were expensive, unpredictable, and potentially ruinous.

The Dutch East India Company (Vereenigde Oost-Indische Compagnie, or VOC), founded in 1602, was one of the earliest and most important experiments with a new structure: a joint-stock company. The VOC was chartered by the Dutch state to establish trade routes to Asia. Here are the key features:

1. **Transferable shares**: Rather than a fixed partnership, the VOC issued shares that could be bought and sold. This allowed capital to be pooled from many investors without requiring them to know each other.
2. **Limited liability**: An investor in the VOC could lose their investment, but not more. Their personal assets were protected. If the VOC went bankrupt, shareholders lost their money; creditors might not be paid in full; but shareholders' homes and fortunes were not at risk.
3. **Separate legal personality**: The VOC was recognized as a legal entity distinct from its owners. It could own property, enter contracts, and sue and be sued in its own name.
4. **Transferable management**: Capital and management were separated. Investors did not have to run the company themselves; professional management could be hired.

The VOC was, undoubtedly, extractive. It used force to establish monopolies on spice trade. It displaced local populations. But it was also *exploratory*: it mounted expensive, risky voyages to discover and open new trade routes. And it was massively profitable, generating returns that attracted investors across Europe. The joint-stock company structure spread to other colonial ventures and eventually to other sectors: banking, insurance, manufacturing, utilities.

By the 19th century, the joint-stock company with limited liability was becoming the dominant form of large-scale enterprise in the developed world. Marx could observe its rise, though he could not fully anticipate its consequences.

2.2 Firm Activity as a Call Option

Here is the key insight from financial theory: limited liability is mathematically equivalent to a *call option*.

Let us use a simple example. Suppose you are considering an investment in a risky venture. The venture requires an initial investment of \$1 million. There are two possible outcomes:

- With probability 40%, the venture succeeds and generates revenues of \$5 million (profit of \$4 million).
- With probability 60%, the venture fails and generates revenues of \$0 (loss of \$1 million).

Expected payoff: $0.4 \times 4 - 0.6 \times 1 = 1.6 - 0.6 = \$1$ million.

Now, here is the crucial question: *under what liability regime would you take this bet?*

Under **unlimited liability** (partnership structure): If you invest \$1 million and the venture fails, you lose the full \$1 million. But if the venture was financed partly with borrowed money—say, you put in \$1 million and borrowed another \$4 million—then failure means you owe the creditors the \$4 million and lose your own \$1 million. You are liable for the full \$5 million loss. You would be reluctant to take a bet with downside risk this large, even if the expected payoff is positive.

Under **limited liability** (corporation structure): If the corporation goes bankrupt, you lose your \$1 million investment and nothing more. The creditors may take a loss; the corporation may be liquidated; but your personal assets are protected. Your maximum loss is the \$1 million you invested.

This is the payoff of a *call option*. You have the right, but not the obligation, to participate in the upside. Your downside is capped. Let us formalize this.

The Call Option Payoff

A call option is a financial contract that gives the holder the right to buy an underlying asset at a specified price (the "strike price" or exercise price), on or before a specified date.

Payoff at maturity:

$$C = \max(S_T - K, 0) \tag{2}$$

where S_T is the price of the underlying asset at time T, and K is the strike price.

Example 1: Call Option on a Stock. You buy a call option on a stock. The current stock price is \$20. The call option has a strike price of \$10 and expires in one year. At expiration:

- If the stock price is \$30, you exercise the option, buy the stock for \$10, and immediately sell it for \$30. Your payoff is \$20.
- If the stock price is \$5, you do not exercise the option. Your payoff is \$0. You do not lose more than the option premium you paid upfront.

Figure 2 shows the payoff diagram for this call option.

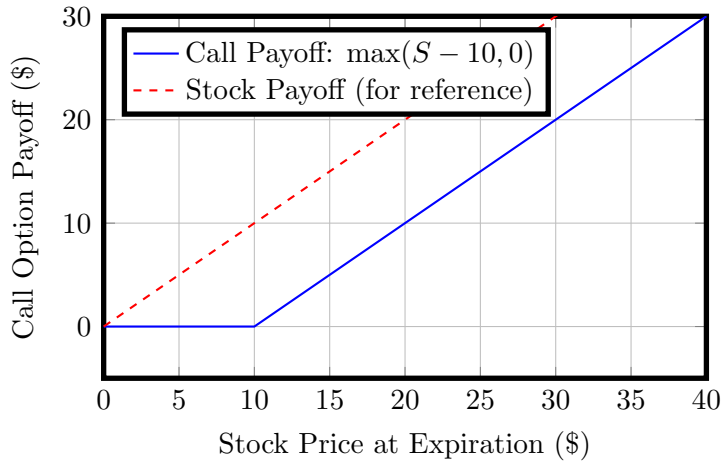


Figure 2: Call Option Payoff: Strike Price = \$10

The key feature: unlimited upside, limited downside (zero payoff below the strike price).

2.3 Equity as a Call Option on Assets

Now consider the equity stake in a limited-liability corporation. Here is a simple balance sheet:

Assets		Liabilities	
Assets	\$20 million	Debt	\$10 million
		Equity	\$10 million
Total	\$20 million	Total	\$20 million

The firm has \$20 million in assets, financed by \$10 million of debt and \$10 million of equity. Now, the key question: what happens to the equity holders if the asset value changes?

- If assets are worth \$20 million: Debt is repaid (\$10 million). Equity holders get the remaining \$10 million. Equity value = \$10 million.
- If assets are worth \$15 million: Debt is repaid (\$10 million). Equity holders get the remaining \$5 million. Equity value = \$5 million.
- If assets are worth \$8 million: Debt claims \$10 million, but assets are only \$8 million. Creditors take a \$2 million loss. Equity holders get nothing. Equity value = \$0.
- If assets are worth \$0: Both debt and equity are worthless. But equity holders do not owe anyone money; they lose at most their \$10 million investment.

The equity stake has a payoff profile identical to a call option. The strike price is the amount of debt (\$10 million). You have unlimited upside if the venture is successful (assets rise above \$10 million debt). You have limited downside: you lose at most your equity investment (\$10 million).

Figure 3 shows the payoff diagram for equity as a function of asset value.

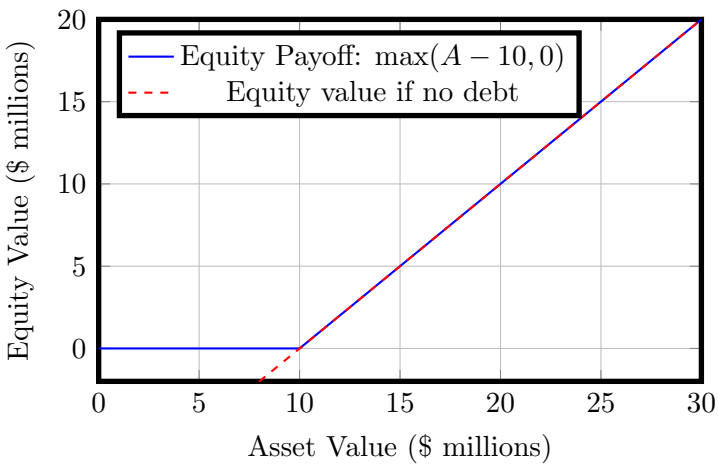


Figure 3: Equity as a Call Option on Assets: Strike Price (Debt) = \$10 million

This structure creates powerful incentives. Let us think through the implications.

2.4 Incentives Under Limited Liability

First, equity holders **like volatility**. Suppose two ventures have the same expected value but different distributions of outcomes:

	Outcome 1 Prob.	Outcome 1 Value	Outcome 2 Prob.	Outcome 2 Value
Venture A (low volatility)	0.5	\$2M	0.5	\$2M
Venture B (high volatility)	0.5	\$5M	0.5	-\$1M

Expected value: Both ventures have an expected payoff of \$2M.

But under limited liability with \$1M equity and \$1M debt:

- Venture A: In both states, revenues are \$2M. Debt is repaid (\$1M). Equity holders get \$1M. Certain payoff to equity: \$1M.
- Venture B:
  - State 1 (prob. 0.5): Revenues \$5M. After debt repayment (\$1M), equity gets \$4M.
  - State 2 (prob. 0.5): Revenues -\$1M. The venture is insolvent. Creditors absorb the loss. Equity gets \$0.

  Expected payoff to equity: $0.5 \times 4 + 0.5 \times 0 = \$2M$.

Wait—both ventures have the same expected value for equity holders. But Venture B is riskier. Why would equity holders prefer higher volatility?

The answer lies in *leverage and the asymmetry of outcomes*. If we lever up even more (say, \$1M equity, \$4M debt, \$5M total capital):

- Venture A (low vol): Revenues \$2M. Debt \$4M. Equity is underwater. Equity gets \$0.
- Venture B (high vol):
  - State 1: Revenues \$5M. After debt \$4M, equity gets \$1M. Return on \$1M equity: 100%.
  - State 2: Revenues -\$1M. Equity gets \$0. Loss is capped at equity investment.

In the leveraged case, the high-volatility venture offers equity holders a chance at large gains, while losses are capped. Low-volatility ventures offer equity nothing (since debt takes priority). This is the *option-like feature of equity*: as leverage increases, risky bets become more attractive to equity holders.

Second, equity holders like new markets and exploration. Exploration and innovation are inherently high-volatility activities. The outcome is uncertain. Some expeditions fail entirely; some discover valuable new resources or trade routes. Under unlimited liability, would you finance an expedition to find a Northwest Passage? The expected return might be high, but the downside risk of losing everything is immense. Under limited liability, you get the upside if the expedition succeeds, but your loss is capped if it fails.

Third, equity holders like to delegate to entrepreneurs. Separation of ownership from management is crucial. If you own shares in the VOC, you do not have to personally command a ship to Asia. Professional captains, merchants, and administrators do the work. This

allows specialized expertise and (relatively) efficient decision-making. More importantly, it allows capital from many small investors to be pooled for ventures that would be too large or risky for any one individual.

Fourth, transfer of shares increases dynamism. Because shares are tradeable, investors are not locked into a venture. If you believe the managers are performing poorly, you can sell your shares (to someone else who is more optimistic). This creates a discipline on management and facilitates capital reallocation.

In sum: limited liability creates an incentive structure that favors risk-taking, exploration, and innovation. It also creates the legal and organizational structure needed to pool capital from many dispersed investors. This is powerful.

2.5 Exploration and the Creation of Value

But Marx might ask: is this really value creation, or just a redistribution? If the VOC discovers a new trade route and accumulates wealth, is that new value, or is it just extraction—shifting wealth from one group (say, indigenous populations or traditional trade monopolists) to another?

The answer is subtly both and neither. Some VOC wealth was indeed extractive, based on force and appropriation. But not all growth comes from moving existing value around.

Here is the key insight: ideas and techniques are *non-rival*. A rival good (like land or labor) can only be used by one person at a time. If I take your wheat, you no longer have it. But an idea (like a new sailing route, a new manufacturing technique, a new form of organization) can be used by many people simultaneously without diminishing the original user's benefit.

When the VOC discovered a faster sailing route to Asia, that knowledge could be (and eventually was) diffused to competitors, complementary enterprises, and subsequent generations. The route was not depleted by use. This is radically different from extracting gold, which is a finite resource.

Again, the VOC had exploitative properties, and if you want a terrifying read of that, see Ghosh (2021). Marx was right. But there is an element of the VOC and the financial innovation of the Dutch that is beneficial for the purpose of non-extractive *exploration* as well.

2.6 The Solow Model and the Residual

Economic growth, in the long run, comes from the accumulation of these non-rival ideas and techniques. This is captured in the Solow growth model, though not always recognized clearly.

A standard Cobb-Douglas production function is:

$$Y_t = A_t K_t^{\alpha} L_t^{1-\alpha} \tag{3}$$

where Y_t is output, K_t is capital, L_t is labor, α is the capital elasticity (typically around 0.3), and A_t is the "total factor productivity" or "Solow residual."

If we assume competitive markets and constant returns to scale, we can solve for the growth rate:

$$\frac{\dot{Y}}{Y} = \frac{\dot{A}}{A} + \alpha \frac{\dot{K}}{K} + (1-\alpha)\frac{\dot{L}}{L} \tag{4}$$

Now here is the puzzle: when Solow computed this for the United States in the mid-20th century, he found that about 87% of growth could not be explained by increases in capital and labor. The vast majority of growth was attributed to $\frac{\dot{A}}{A}$, the growth in total factor productivity.

This is the "Solow residual"—the part of growth that we cannot explain by simply working harder (more labor) or accumulating more capital (more machinery, buildings, infrastructure). It is growth that comes from doing things *better*: better techniques, better organization, better ideas, better knowledge.

Figure 4 illustrates this for the United States. The blue line shows actual GDP growth rates over time. The red dashed line shows growth that would be predicted by capital accumulation and labor force growth alone (using the Cobb-Douglas model with $\alpha = 0.3$). The gap between the two is the Solow residual—the contribution of technological progress and efficiency improvements.

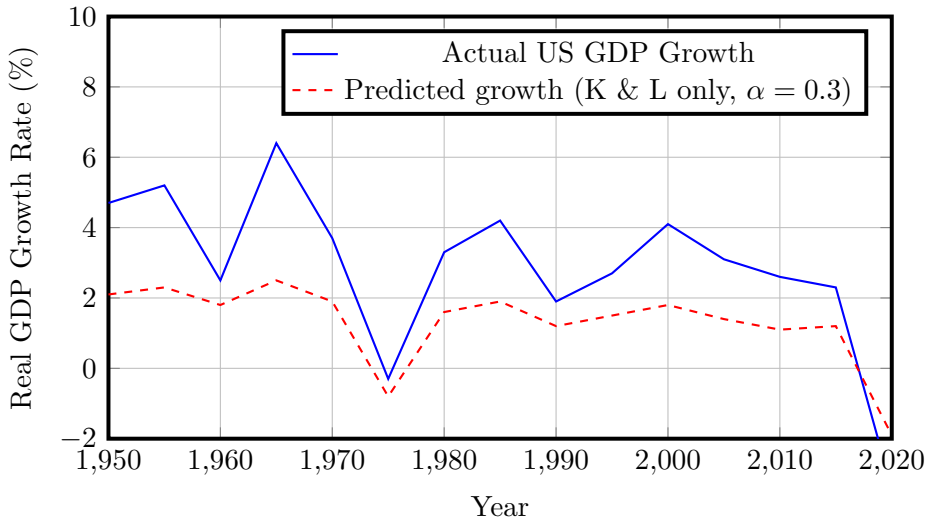


Figure 4: US GDP Growth: Actual vs. Predicted by Capital and Labor. The gap is the Solow Residual (TFP growth).

The persistent gap shows that something besides capital and labor is driving growth. That

"something" is the Solow residual.

To obtain the Solow residual, society seems to benefit from a sustained incentive to reward innovation and risk-taking by entrepreneurs. The limited-liability corporation, combined with equity markets and the profit motive, provides exactly that incentive.

The incentive to explore and innovate creates a stream of new techniques and new goods that would not exist otherwise. The smartphone would not exist if no one had an incentive to invest in research and development. Antibiotics would not be manufactured at low cost if pharmaceutical companies had no prospect of profit. The incentive structure matters for what gets invented and how widely it diffuses.

Marx understood that capitalism was dynamic, innovative, and constantly revolutionizing production. But he interpreted this as ultimately destructive: capitalism would eventually exhaust its possibilities, concentrate wealth to the breaking point, and collapse in crisis. What Marx could not foresee—what required the subsequent development of financial theory and economic data—was that the capitalist incentive structure would generate enough continuous innovation to sustain growth and raise living standards, even as it concentrated wealth.

This is not an argument that capitalism is perfect, or that it distributes benefits fairly, or that it has no downsides. It is an argument that one crucial source of growth—the Solow residual, the accumulation of ideas through exploration and innovation—depends on a specific legal and financial structure: the limited-liability corporation with tradable equity and profit incentives.

Conclusion

The modern mode of production works through the limited-liability corporation that creates powerful incentives for risk-taking, exploration, and innovation. Combined with the ability to accumulate capital from dispersed investors and to transfer ownership easily, this structure has enabled the funding and development of ventures that no individual or partnership could undertake: railroads, electricity grids, pharmaceutical research, internet infrastructure, etc.

Further Reading

- On the VOC and early joint-stock companies: Carter & Divine (2016), Gelderblom (2013)
- On limited liability and incentives: Jensen & Meckling (1976)
- On the Solow residual and growth: Solow (1956), Romer (1990)
- On historical living standards and development: Rosling (2018), Pomeranz (2000)

References

Gapminder (2015). Child Mortality Rate, under age five, version 7. Available at: `https://www.gapminder.org/data/documentation/gd005/`

UN Inter-agency Group for Child Mortality Estimation (2024). Child mortality estimates. Available at: `https://childmortality.org/`

Carter, S. & Divine, D. (2016). The historiography of the Dutch East India Company. *Journal of Economic History*, 76(4), 1012–1039.

Gelderblom, O. (2013). *The Political Economy of the Dutch East India Company.* Oxford University Press.

Ghosh, A. (2021), *The Nutmeg's Curse: Parables for a Planet in Crisis.* Chicago: The University of Chicago Press.

Jensen, M. C. & Meckling, W. H. (1976). Theory of the firm: Managerial behavior, agency costs and ownership structure. *Journal of Financial Economics*, 3(4), 305–360.

Solow, R. M. (1956). A contribution to the theory of economic growth. *The Quarterly Journal of Economics*, 70(1), 65–94.

Romer, P. M. (1990). Endogenous technological change. *Journal of Political Economy*, 98(5), S71–S102.

Rosling, H., Zhang, O., & Rosling, A. (2018). *Factfulness: Ten Reasons We're Wrong About the World—and Why Things Are Better Than You Think.* Flatiron Books.

Pomeranz, K. (2000). *The Great Divergence: China, Europe, and the Making of the Modern World Economy.* Princeton University Press.

3 The power of diversification (CAPM)

3.1 The Problem: Risk Remains

But where does the risk go? The risk on the asset side shifts to equity and debt holders. If a venture is genuinely risky, someone must bear that risk. The question is: who bears it, and how much do they demand in compensation?

Under unlimited liability (the old partnership model), the entrepreneur bears the full risk. If you mount an expedition and it fails, you lose everything—your home, your assets, your family's security. This creates a powerful incentive to be conservative, to avoid bold bets. Growth slows.

Under limited liability, the equity holders bear the risk of the venture. But equity holders are not just the bold entrepreneurs; they are dispersed investors with savings they want to grow. These investors are typically risk-averse. They would not invest in a single risky venture; they would demand an enormous return to compensate for the risk of total loss.

But here is where finance provides its magic: *diversification*. If you own a small stake in many ventures—some succeed, some fail, on average you earn a decent return—you can dramatically reduce the risk you bear. The key insight is that most of the risk in any single venture is *idiosyncratic*: specific to that venture. If you diversify, these idiosyncratic risks cancel out. What remains is *systematic risk*: the risk that is common to all ventures, correlated with the overall economy.

Finance theory says: investors should only be compensated for systematic risk. Idiosyncratic risk can be diversified away for free. This is the power of diversification, and it is what allows risky ventures to get funded at reasonable rates.

3.2 The SpaceX Example

Consider SpaceX, Elon Musk's rocket company. SpaceX is unquestionably a high-risk venture. Rockets explode. Development timelines slip. Competition is intense. The company's success depends on solving extraordinarily difficult engineering and manufacturing problems.

Yet in 2021, the Financial Times reported: "Another advantage that SpaceX has shared with Tesla has been its ready access to cheap capital, thanks to the high valuation investors have been prepared to put on its business."[2] SpaceX was able to raise capital at valuations that implied low cost of capital, despite the high risk of the individual project.

How is this possible? Why would investors provide cheap capital to a company whose rockets regularly fail?

The answer: diversification. SpaceX is a single venture, but investors in SpaceX are not putting all their wealth into SpaceX. Venture capital investors diversify across a portfolio of companies. Some will fail spectacularly. Others will succeed wildly. On average, if the portfolio is well-constructed, returns are reasonable and risk is manageable.

From SpaceX's perspective, the cost of capital it faces depends not on the volatility of its own rockets, but on how its returns correlate with the overall market. If SpaceX's fortunes are uncorrelated with (or negatively correlated with) the broader economy, then from a diversified investor's perspective, SpaceX is actually less risky than it appears in isolation. It provides a hedge.

This is the insight of the Capital Asset Pricing Model (CAPM), one of the foundational concepts in finance.

[2] Financial Times (2021)

3.3 The CAPM: Cost of Capital Depends Only on Beta

The Capital Asset Pricing Model provides a formula for the cost of capital (or required return) on an investment:

$$r_i = r_f + \beta_i(r_m - r_f) \tag{5}$$

where:

- r_i is the required return (cost of capital) for asset i
- r_f is the risk-free rate (return on a safe investment, like a government bond)
- r_m is the expected return on the market portfolio
- β_i is the *beta* of asset i: the sensitivity of its returns to market returns
- $(r_m - r_f)$ is the market risk premium: the extra return investors demand for bearing systematic risk

The formula says: the required return on an investment is the risk-free rate plus a risk premium. The risk premium is proportional to β, not to the total volatility of the investment.

This is the crucial insight. Let us think through why.

3.4 Why Only Beta Matters: The Diversification Argument

Imagine you are a portfolio manager. You hold a large, diversified portfolio of many assets. You are considering adding a new asset—say, SpaceX shares.

The returns on SpaceX are volatile. Quarter to quarter, the stock might swing 20% or more. From a narrow perspective, this is risky. But you do not hold only SpaceX; you hold many assets. Some of SpaceX's volatility is *idiosyncratic*: it is specific to SpaceX and uncorrelated with your other holdings. When SpaceX has a bad quarter, your other holdings might have a good quarter. The volatility cancels out.

What does not cancel out is *systematic risk*: the part of SpaceX's returns that is correlated with your other holdings and with the broader market. If SpaceX's returns move together with the market—when the market is up, SpaceX tends to be up; when the market is down, SpaceX tends to be down—then you cannot diversify this away. This systematic risk is captured by β.

Mathematically, we can decompose the variance of any asset's returns into two parts:

$$\mathrm{Var}(r_i) = \beta_i^2 \sigma_m^2 + \sigma_\epsilon^2 \tag{6}$$

where:

- $\beta_i^2 \sigma_m^2$ is the *systematic variance*: the part driven by market movements
- σ_ϵ^2 is the *idiosyncratic variance*: the part specific to asset i

When you hold a well-diversified portfolio, the idiosyncratic variance terms largely cancel out across assets. But the systematic variance remains. It is this systematic variance that you cannot escape, no matter how much you diversify.

Therefore, investors should only demand compensation for systematic risk (β), not for idiosyncratic risk. Idiosyncratic risk can be eliminated for free, through diversification. Demanding a return premium for something that can be eliminated for free would be irrational.

This is why, according to CAPM, the cost of capital depends only on β and not on the total volatility of the investment.

3.5 The Implication: Low-Beta Projects Get Cheap Capital

Here is the practical implication. Suppose you have two projects:

Project A: High volatility, but low beta. The project has volatile returns, but those returns are uncorrelated with the market. When the market crashes, this project is unaffected (or benefits). When the market booms, this project is unaffected. From a diversified investor's perspective, this project is a hedge. It reduces overall portfolio risk.

Project B: Moderate volatility, high beta. The project's returns are highly correlated with the market. When the market crashes, this project crashes too. When the market booms, this project booms.

According to CAPM, Project A should have a *lower* cost of capital than Project B, even though Project A is more volatile in isolation. Why? Because Project A's volatility is idiosyncratic and can be diversified away. Project B's volatility is systematic and cannot be diversified away.

This seems counterintuitive to someone who has not studied finance. If you told a non-finance person, "This project is very risky, so we will finance it cheaply," they would be baffled. But the logic is sound, given the assumption of well-diversified investors.

Now apply this to SpaceX. SpaceX is a high-volatility venture. Rockets explode; the stock price swings wildly. But SpaceX's fortunes are not perfectly correlated with the overall market. Venture capital investors diversify across many startups. From their perspective, much of

SpaceX's volatility is idiosyncratic. Therefore, SpaceX's beta is lower than its volatility alone would suggest. Therefore, SpaceX's cost of capital is lower than it would be if investors were undiversified.

This is why SpaceX can raise capital cheaply despite high idiosyncratic risk. Finance theory explains it; diversification makes it possible.

(NB: Reality is messier than the CAPM assume. Investors sometimes panic and behave irrationally. Diversification has costs (time, fees). Markets are not always efficient. Taxes and transaction costs do matter. Information is asymmetric. But that's for the Investments Course to deal with.)

Further Reading

- On CAPM and the intuition of beta: Sharpe (1964), Bodie & Marcus (2014)
- On diversification and portfolio theory: Markowitz (1952)
- On venture capital and diversification: Gompers & Bauer (2010)

References

Sharpe, W. F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *The Journal of Finance*, 19(3), 425–442.

Bodie, Z. & Marcus, A. J. (2014). *Investments* (10th ed.). McGraw-Hill Education.

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77–91.

Gompers, P., Lerner, J., & Scharfstein, D. (2010). *Venture Capital and Private Equity: A Casebook* (5th ed.). Wiley.

Financial Times (2021). SpaceX's Cheap Capital Advantage. Retrieved from `https://www.ft.com/content/25e2292b-a910-41c8-9c55-09096895f673`

4 The power of banks (Diamond-Dybvig)

We have explained why equity holders are willing to bear risk: they hold an option on the firm's assets and they can diversify to mitigate the risk in their portfolio. But how about the debt holders?

A farmer saves money for next season's seed. A family sets aside funds for an emergency. A worker wants to borrow to buy a house. These savers and borrowers are small and cannot

diversify; they need access to capital at reasonable rates. And they won't invest in stocks that easily.

Risky projects (like a farmer's harvest, a factory's expansion, a business venture) take time to pay off. They might not return cash for months or years. But savers need access to their money sooner. A family's savings might be needed in a few months for an unexpected expense. A worker wants a mortgage but wants to pay it back over 30 years, not wait for the long-term payoff of the house.

This creates a mismatch: savers want short-term liquidity; productive investments require long-term credit (loans). How is this bridged?

The answer is the bank. Banks solve this problem through *liquidity and maturity transformation.* A bank borrows short-term (takes deposits that savers can withdraw on demand, or at least quickly) and lends long-term (finances risky projects that take years to pay off). As long as not all savers withdraw at once, the bank can manage the timing mismatch and provide both liquidity to savers and financing to borrowers.

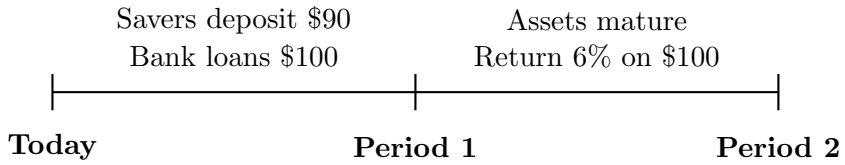
4.1 A Simple Balance Sheet

Consider a stylized bank:

Assets		Liabilities	
Risky loans	100	Deposits (savings)	90
		Equity	10
Total	100	Total	100

The bank has \$100 in assets: loans to farmers, small businesses, and other borrowers. These loans are risky; they might return more or less than expected. The bank finances this with \$90 in deposits (short-term liabilities from savers) and \$10 in equity (the bank's own capital).

Now consider the timeline:



Period 1: Savers have deposited \$90. The bank has lent out \$100 (using \$10 equity as cushion). The bank's borrowers are working on their projects. The bank pays interest on deposits: 2% on \$90 = \$1.80. This comes from the equity cushion temporarily.

Period 2: The risky assets mature. The bank receives a 6% return on its $100 loans = $6. This pays back the deposits plus interest, plus contributes to equity.

If everything goes smoothly:

- Savers get their $90 back plus 2% interest = $91.80
- Bank equity: $10 + ($6 assets return) - ($1.80 interest paid) = $14.20
- Everyone is happy. Savers got safe, liquid savings. Borrowers got long-term financing. Bank earned a spread.

This is the value of banking: it transforms risky, illiquid assets (loans) into safe, liquid liabilities (deposits). As long as not everyone withdraws at once, the system works.

4.2 The Diamond-Dybvig Problem: Liquidity Shocks and Bank Runs

But what if some savers suffer a liquidity shock? They need their money urgently—an illness, a business failure, an opportunity. They want to withdraw, and they do not want to wait for the full return.

Here is the problem: the bank's assets are not easily liquidated. If the bank tries to sell the loans (to farmers, small businesses) before they mature, buyers will demand a discount. The liquidation value is only 80% of face value.

So if the bank needs to liquidate $20 of assets to pay withdrawals, it receives only $16. It loses $4.

Let us think through what happens:

Scenario 1: Small withdrawal demand (say, $30 withdraws early)

- Bank liquidates $30 of assets, receives $24 (80% of $30)
- Loss: $6
- Bank pays out: $30
- Bank's remaining assets: $70 (at par) + $24 (just liquidated) = $94
- But bank has paid $30, so cash position is $24 - $30 = -$6
- Bank must tap equity: equity absorbs the $6 loss
- Equity now: $10 - $6 = $4
- Remaining depositors are still okay. The bank can meet all obligations.

Scenario 2: Large withdrawal demand (say, $60 withdraws early)

- Bank liquidates $60 of assets, receives $48 (80% of $60)
- Loss: $12
- Bank pays out: $60
- Cash received: $48, but must pay $60
- Shortfall: $12
- Remaining assets: $40 (at par) worth $100 (full assets) - $60 (liquidated)
- Bank equity: $10 - $12 = -$2
- Bank is insolvent. Remaining depositors are in trouble.

The critical threshold: the bank is solvent as long as withdrawals do not exceed about $50 million. Why $50M? Because liquidating $50M of assets at 80% = $40 received, so a $10M loss. The bank's equity is $10M, just enough to absorb this loss. But if withdrawals exceed $50M, losses exceed equity, and the bank is bankrupt.

4.3 The Bank Run

Here is the vicious logic. Once savers become aware of this fragility, they have an incentive to withdraw early. If you know that the bank can only survive $50M of withdrawals, and you hear rumors that large withdrawals are happening, you want to be first in line. If you wait, you might get nothing.

This creates a *bank run*: savers rush to withdraw simultaneously. Even a healthy bank—one that would be solvent if savers waited—can fail if a large fraction withdraw at once.

This is the Diamond-Dybvig problem, named after the 1983 paper by Douglas Diamond and Philip Dybvig.[3] They showed that a bank providing valuable maturity transformation (borrowing short, lending long) is inherently fragile. It is vulnerable to self-fulfilling runs: if savers believe the bank will fail, they withdraw, which causes the bank to fail.

This fragility is not a flaw in bank management or individual bank risk-taking. It is a structural feature of maturity transformation itself.

4.4 Regulation and Deposit Insurance

If maturity transformation is valuable but fragile, what can be done?

[3] Diamond & Dybvig (1983)

The standard solution, implemented in most developed economies, is *regulation and deposit insurance*:

1. **Deposit Insurance**: A government agency (in the US, the FDIC) guarantees that deposits up to a certain level (currently $250,000 in the US) are protected, even if the bank fails. This removes the incentive for depositors to run. You do not need to rush to withdraw because your deposit is safe.
2. **Capital Requirements**: Banks must maintain a minimum ratio of equity to assets (e.g., 8-10%). This provides a cushion to absorb losses from illiquid assets. The higher the equity cushion, the more withdrawals a bank can absorb before insolvency.
3. **Liquidity Requirements**: Banks must maintain a minimum amount of liquid assets (cash, short-term securities) relative to deposits. This ensures the bank can meet normal withdrawal demand without liquidating risky assets at steep discounts.
4. **Supervisory Oversight**: Regulators monitor banks' balance sheets, lending practices, and risk exposures. If a bank is taking on too much risk, regulators can force corrective action before insolvency.

These regulations solve the fragility problem. Deposit insurance eliminates the incentive for runs (depositors know they are protected). Capital and liquidity requirements reduce the likelihood of insolvency. Together, they make maturity transformation stable and sustainable.

The irony is that without regulation, banks cannot function well. With regulation, they can provide valuable intermediation: transforming risky, long-term assets into safe, liquid deposits. Households and firms can borrow and save at reasonable rates. Growth is supported.

4.5 From Fragility to Financial Architecture

The regulatory structure we observe today—deposit insurance, capital requirements, liquidity standards, central banks as lenders of last resort—is largely a response to the maturity transformation problem that Diamond and Dybvig formalized.

This matters for understanding modern capitalism. Banks are not just intermediaries moving money around. They are essential institutions that solve the timing and risk-bearing problems created by the mismatch between savers' and borrowers' horizons. Without banks, savers would demand very short-term, liquid returns (because they need flexibility), and borrowers would have to pay very high rates (because they have long, uncertain timelines). The economy would be starved of long-term financing.

But banks cannot exist stably without the regulatory framework that prevents runs. This is why we have central banks, deposit insurance, and capital requirements. These are not unnecessary bureaucratic constraints; they are solutions to a fundamental problem of financial intermediation.

Marx could observe that capitalism concentrates wealth and creates financial crises. What Marx could not fully theorize is that capitalism also develops solutions to its own fragilities: legal innovations (limited liability, joint-stock companies), financial innovations (diversification, derivatives), and regulatory innovations (central banking, deposit insurance). The system is not static; it evolves to solve the coordination and risk-bearing problems that arise.

References

Diamond, D. W. & Dybvig, P. H. (1983). Bank runs, deposit insurance, and liquidity. *The Journal of Political Economy*, 91(3), 401–419.

5 Conclusion

The Finance view helps us to understand four phenomena of a modern economy with high living standards:

1. How value is created through capital budgeting.
2. How limited liability companies are uniquely positioned to take risks and innovate.
3. The power of diversification, that makes it possible for risky companies and start-ups to find financing.
4. How banks provide liquidity and funding. And why the financial system is so heavily regulated.

It does lead to new questions:

- Developing countries usually have a very small financial sector. Is that a cause or an effect?
- How do we know that rich countries are not exploiting poor countries? Or that growth exploits nature?
- Are there alternative systems that would make the world better off?
- Can growth continue indefinitely or are there limits to growth?
- Are people in modern economies not suffering from fatigue and stress as a by-product of technological development? Is all that growth real?
- What is the proper role of government?
- Can the Finance view be applied to any activity in the economy or are there some limits? Are there moral limits?

- What are the pre-conditions for Finance to contribute to social welfare?

Such questions could be raised in the discussion groups. Or we might address a selection of it in the next lectures.